

1. Permitted URL input

Preserved source identity; declared parsing and exclusions
External input: publisher url_norm under disclosed amendment

2. Structural inference

25 structural features + fixed Logistic-L1 model
Validation-selected threshold; no test-time refit

3a. Fixed cascade

Outside fixed band: Logistic-L1 decision
Inside fixed band: character transformer
Separate fixed transformer threshold
Allow/alert output; no automatic blocking

3b. GMM monitor

25 features + first-stage score
256-row windows; stride 64
Strict frozen alert boundary

4. Future-only policy comparator

An alert ending at request t
escalates $t+1$ through $t+256$
Never revises the triggering window

5a. Detection and policy evaluation

Save paired scores, decisions and routes
Then join source/tier outcome strata
Domain-clustered recall contrasts
Low-FPR gates and secondary metrics
Outcome labels do not drive routing

5b. Separate real-HTTP measurement

125 cells / 25 five-repeat groups
Actual physical transformer attempts
Terminal latencies and request errors
Client-phase throughput and drain
No new model or threshold selection

Dashed path: override for future requests only.